

GROUP MEMBERS :

- 1 : Akhil Prajapati
- 2 : Smit Panchal
- 3 : Shiv Jadhav
- 4 : Smit Thakor

AGENDA

A Hybrid E-bike is a bicycle equipped with both pedal power and an electric motor for assistance.

It can be operated in three modes: pedal-only, motor-only, or pedal-assist mode.

The battery is rechargeable using a normal electricity supply, and some models also support solar charging.

Hybrid E-bikes are widely used in cities, campuses, and short-distance travel as an alternative to petrol-based vehicles.

They provide benefits like zero emissions, cost savings, and fitness improvement.

Common applications include: daily commuting, delivery services, eco-tourism, fitness rides, and reducing traffic congestion.

BENIFITS:-

1. Eco-Friendly – Low carbon emissions
2. Affordable – Reduces travel costs
3. Healthy – Fitness with convenience
4. Urban Mobility – Easy commuting & less traffic load

IDEATHON 4.0

HYBRID E BIKE



KEY FEATURS

1. Dual Power Source: Pedal + Electric Motor
2. Rechargeable Lithium-Ion Battery
3. Multiple Riding Modes (Eco, Normal, Sport)
4. Regenerative Braking System
5. Adjustable Speed Control
6. Digital Display for Battery & Speed Monitorinm
7. Long Riding Range (30–80 km per charge)
8. Lightweight and Portable Design
9. Low Maintenance
10. Environment Friendly

FUTURE SCOPS

- 1.Solar charging integration
2. IoT & AI-assisted riding
3. Advanced safety features
4. Wider adoption for smart cities

